ZIBANG NIE

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Research Interests: Computer Vision, Natural Language Processing
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EDUCATION

Central South University (Project 985) **Changsha, China**
B.Eng. in Computer Science & Technology *Sep 2022 - Jun 2026 (Expected)*
·GPA:3.7/4.0 | Average Score: 87/100

University of Dundee (Dual-Degree Program) **Dundee, UK**
B.Sc. (Hons.) in Computing Science *Sep 2022 - Jun 2026 (Expected)*
·GPA:3.7/4.0 | Average Score: 87/100

Rice University **Houston, USA**
Master of Computer Science *Aug 2026 - Dec 2027 (Expected)*

PUBLICATIONS & PATENTS

Journal/Conference:

Zhong, Y., Tang, Z., Zhang, H., Dai, Z., ***Nie, Z.***, Xie, Y. "Pulp Grade Monitoring Using Binocular Image Through Multi-Scale Feature Cross-Attention Fusion Network and Saliency Map Constraint," Advanced Engineering Informatics, 2024. DOI: 10.1016/j.aei.2024.102780. (SCI Q1, IF≈6.1)

Luo, J., Tang, Z., Zhang, H., Fan, Y., Xie, Y., ***Nie, Z.***, Gui, W. "A Defocused Calibration Method Using ROI-Patched MAM-UNet and Progressive Training for Froth Flotation Process," IEEE/ASME Transactions on Mechatronics, 2025. DOI: 10.1109/TMECH.2025.3545743. (SCI Q1, IF≈6.8)

Nie, Z., Fan, Y., He, F. "Robustness Analysis of Visual Transformer Based on Adversarial Attacks," Applied and Computational Engineering, 2024. DOI: 10.54254/2755-2721/41/20230737.

Luo, H., Tang, Z., Zhang, H., Xie, Y., ***Nie, Z.*** "TS-DDPD: A Transfer Strategy with Differential Distribution and Progressive Distillation for Working Condition Monitoring in Froth Flotation Process," SSRN Preprint, 2025. DOI: 10.2139/ssrn.5184198. (Under Review)

Patents:

Tang, C., Ma, W., Zhang, H., Su, Y., ***Nie, Z.*** "Data-Fusion-Based Method for Building a Flotation Key Indicator Monitoring Model." China National Invention Patent Application, Pub. No. CN118887587A, Published 2024.

Zhang, H., Tang, C., Ma, W., Tang, H., ***Nie, Z.*** "Foam-Video Key-Frame Extraction Method Based on Average Foam Velocity." China National Invention Patent Application, Pub. No. CN118887586A, Published 2024.

RESEARCH & PROJECT

Efficient Pipeline-Parallel Training & ViT Adversarial Robustness

Research Assistant *Jun 2023 - Sep 2023*

Reproduced vPipe on multiple GPUs using swap, recomputation, and dynamic repartitioning to reduce activation memory pressure and improve pipeline efficiency for large-scale Transformer training. I also evaluated adversarial robustness on CIFAR-10 with FGSM and I-FGSM, comparing ViT-Base with ResNet-18. The project achieved about 2× throughput over PipeDream, enabled training of models 1.2× larger, and showed that ViT suffered 50% less accuracy drop than CNNs at $\varepsilon = 0.3$. Research results were published in Applied and Computational Engineering (2024).

Enhancing Large Language Models with Specialized Knowledge Base

Research Assistant *Dec 2023 - Apr 2024*

Developed a plugin that integrated a specialized local knowledge base with Baidu's Wenxin Yiyan LLM to improve domain-specific question answering in fields such as medicine, law, and science. Built with LangChain, the system combined text embedding, vector similarity search, and customized response generation to improve both answer accuracy and domain expertise. It was tested on over 10,000 queries, integrated into educational platforms and research tools, and outperformed standard LLMs in subjects such as physics and law. The project also received several provincial-level awards in university student competitions.

Deep Learning for Natural Language Processing

Student Researcher *Mar 2024 - Jun 2024*

Built an end-to-end NLP pipeline for ad-tech and fintech datasets, covering raw text cleaning, topic classification, and text similarity prediction. Implemented tokenization, stop-word removal, stemming, TF-IDF vectorization, PCA dimensionality reduction, and Jaccard/cosine similarity metrics in Python, and packaged the workflow into reusable functions in utils.py to process more than 10,000 documents. The pipeline extracted top keywords for each topic and achieved a macro-F1 of 0.82 on unseen data. The project was graded 92/100 and received a completion certificate signed by Patrick Houlihan of Columbia University.

Industrial Froth-Flotation Field Research, Fan-kou Lead-Zinc Mine

Research Assistant *Jul 2024 – Present*

Conducted industrial field research on froth-flotation imaging at the Fan-kou Lead-Zinc Mine, where I deployed a binocular imaging setup at rougher cells, synchronized image streams with DCS tags, operator logs, and lab assays, and curated around 8,000 high-resolution froth images. I supported three related studies: pulp-grade prediction using multi-scale cross-attention fusion with a Video Transformer backbone, defocus-robust calibration using ROI-Patched MAM-UNet with progressive training, and cross-condition monitoring using TS-DDPD transfer and distillation on real zinc rougher data. This work contributed to papers in AEI 2024 and IEEE/ASME T-Mech 2025, a 2025 preprint, and two Chinese invention patents (CN118887586A; CN118887587A).

Self-Learning Adaptive Chunker for Long-Document RAG

Student Researcher *Oct 2025 – Apr 2026*

Led the design and implementation of Self-Learning Adaptive Chunker (SLAC) for structured long documents such as standards, regulations, and technical manuals, building an end-to-end RAG pipeline covering chunking, retrieval, reranking, evidence organization, and answer generation. The project introduced structure-aware chunking, boundary refinement, and multi-granularity evidence retrieval to improve the accuracy, faithfulness, and verifiability of long-document question answering while preserving document hierarchy. I also developed document preprocessing, index construction, online retrieval, reranker-based ranking, evidence packing, and OpenWebUI integration to support interactive use and debugging on real document collections.

HONORS & AWARDS

Academic Honors

Third-Class Scholarship, Central South University | 2023
University "Outstanding Student", Central South University | 2023

Academic Competitions

- Provincial Third Prize (Top 20%), 15th China College Students Service-Outsourcing Innovation & Entrepreneurship Competition | 2024
- Provincial Third Prize (Top 15%), China International College Students "Internet+" Innovation & Entrepreneurship Competition | 2024
- Provincial Grant (Top 20 %), National Undergraduate Innovation & Entrepreneurship Training Program | 2024

SKILLS

Languages:

Chinese (Native), English (TOEFL 110: Reading30, Listening28, Speaking23, Writing29)

Technical Skills：

- Programming Languages: Python · C++ · Java · MATLAB
- Deep-Learning Frameworks: PyTorch · TensorFlow · Hugging Face Transformers · CUDA
- Data / CV Libraries: NumPy · Pandas · OpenCV
- Dev Tools: Git · Docker · Linux (Ubuntu)

Selected Coursework (NLP/ML-relevant): Algorithms & AI; Theory of Computation; Mathematical Logic; Probability & Statistics; Multi-paradigm Programming & Data Structures; Computer Systems (2A/2B); Database Systems; Networks & Data Communications.